Modelado de Datos Autónomo: Sistemas Multi-Agente para Normalización 3NF y Generación de Esquemas

Autonomous Data Modeling: Multi-Agent Systems for 3NF Normalization and Schema Generation

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***Index Terms*—Palabras clave:** **agentes autónomos, sistemas multi-agente, normalización de bases de datos, tercera forma normal (3FN), generación de esquemas, diccionario de datos, modelos de lenguaje grandes (LLM), PostgreSQL.**

Keywords: **autonomous agents, multi-agent systems, database normalization, third normal form (3NF), schema generation, data dictionary, large language models (LLMs), PostgreSQL.**

I. Introduction

El diseño de bases de datos relacionales consiste en traducir requisitos de negocio, habitualmente expresados en lenguaje natural, en un esquema físico capaz de soportar la operación de un sistema con garantías de integridad, consistencia y desempeño. Los cimientos de este proceso fueron establecidos por Codd con el modelo relacional [1] y consolidados mediante la definición formal de las dependencias funcionales y de las formas normales [2], mientras que el modelo entidad-relación [3] aportó el puente semántico entre el mundo real y la estructura lógica. Bajo este marco, alcanzar la Tercera Forma Normal (3FN) no es un objetivo cosmético: supone la eliminación de las anomalías de inserción, actualización y borrado, y permite construir diccionarios de datos coherentes y trazables.

No obstante, la práctica profesional dista mucho de este ideal. Desde la toma de requisitos hasta la redacción del código DDL, el diseñador debe identificar entidades y atributos, determinar dependencias funcionales, detectar dependencias transitivas, descomponer relaciones y documentar cada columna en un diccionario de datos. Cada una de estas etapas es vulnerable al error humano, y las herramientas CASE comerciales, como TOAD Data Modeler, asisten la representación gráfica y la generación parcial de código, pero no automatizan la decisión semántica que garantiza la normalización. En especificaciones complejas —con decenas de entidades, claves compuestas y cadenas de dependencias transitivas—, la carga cognitiva y la presión de plazos producen esquemas que incumplen la 3FN, acumulan redundancia disfrazada de rendimiento y generan diccionarios incoherentes, cuyo costo se posterga a migraciones y correcciones en producción.

La aparición de los modelos de lenguaje grandes (LLM) ha abierto una vía prometedora para automatizar este ciclo de vida. Trabajos recientes han mostrado que los LLM pueden derivar modelos conceptuales y esquemas preliminares a partir de especificaciones en lenguaje natural [4], y que arquitecturas multi-agente son capaces de generar esquemas relacionales de alta calidad desde requisitos [5]. Al mismo tiempo, la literatura evidencia límites claros: Kohita [6] demostró que el nivel de normalización del esquema modifica de forma sistemática la precisión con que los LLM generan SQL; Kanchinadam *et al.* [7] mostraron que variantes estructurales del mismo modelo conceptual producen respuestas divergentes en tareas Text-to-SQL; y ERBench [8] evidenció que los LLM cometen errores lógicos al deducir relaciones entre entidades, incluso en esquemas sencillos. Estos resultados convergen en una conclusión: la generación confiable de esquemas no puede delegarse en un único prompt, sino que requiere una verificación explícita y continua de las dependencias funcionales.

En el ámbito de las arquitecturas multi-agente, la evidencia contemporánea confirma que la especialización de roles reduce

los errores lógicos. AttributeForge [9] orquesta 43 agentes especializados para modelar esquemas de producto en un entorno de producción, mientras que GalaxyWeaver [10] demuestra que un ecosistema de agentes puede evaluar iterativamente un esquema deficiente y reestructurarlo de forma autónoma hasta alcanzar un estado óptimo. Estos antecedentes sugieren que la misma estrategia puede aplicarse al problema de la normalización relacional, siempre que exista un agente dedicado a auditar las dependencias matemáticas que sostienen el esquema.

En este artículo se presenta un framework empírico de modelado de datos autónomo compuesto por agentes especializados: un agente analista extrae entidades, atributos y relaciones a partir del texto de requisitos; un agente generador construye el diccionario de datos; un agente codificador redacta el código DDL para PostgreSQL; y un agente orquestador evalúa las dependencias funcionales e itera sobre el esquema hasta demostrar su conformidad con la 3FN antes de emitir el DDL final. Las contribuciones del artículo son: (i) la definición de una arquitectura multi-agente con roles cognitivos separados para extracción de requisitos, documentación y codificación, auditada por un verificador de dependencias; (ii) un protocolo de verificación que opera bajo los criterios formales de normalización de Codd [2] como métrica de éxito; (iii) una evaluación empírica sobre especificaciones complejas reales que mide la tasa de éxito del sistema para producir esquemas en 3FN libres de anomalías; y (iv) una comparación cuantitativa del tiempo de diseño frente al proceso manual con TOAD Data Modeler.

El resto del artículo se organiza como sigue. La Sección II presenta el marco teórico de normalización sobre el que se apoya el sistema. La Sección III describe la arquitectura del framework y los protocolos de interacción entre agentes. La Sección IV detalla el protocolo experimental y los casos de prueba. La Sección V analiza los resultados obtenidos y la Sección VI resume las conclusiones.

English Version

The design of relational databases consists of translating business requirements, usually expressed in natural language, into a physical schema able to support system operation with guarantees of integrity, consistency, and performance. The foundations of this process were established by Codd with the relational model [1] and consolidated through the formal definition of functional dependencies and normal forms [2], while the entity-relationship model [3] provided the semantic bridge between the real world and the logical structure. Within this framework, achieving the Third Normal Form (3NF) is not a cosmetic goal: it eliminates insertion, update, and deletion anomalies and enables the construction of coherent, traceable data dictionaries.

In practice, however, professional workflows fall short of this ideal. From requirements gathering to DDL code authoring, the designer must identify entities and attributes, determine functional dependencies, detect transitive dependencies, decompose relations, and document every column in a data dictionary. Each of these stages is prone to human error, and commercial CASE tools such as TOAD Data Modeler assist graphical representation and partial code generation but do not automate the semantic decision that guarantees normalization. In complex specifications —with dozens of entities, composite keys, and chains of transitive dependencies— cognitive load and schedule pressure yield schemas that violate 3NF, accumulate redundancy disguised as performance, and produce incoherent data dictionaries whose cost is deferred to production-time migrations and fixes.

The emergence of large language models (LLMs) has opened a promising avenue for automating this life cycle. Recent work has shown that LLMs can derive conceptual models and preliminary schemas from natural-language specifications [4], and that multi-agent architectures can generate high-quality relational schemas from requirements [5]. At the same time, the literature documents clear limits: Kohita [6] showed that the normalization level of a schema systematically affects the accuracy with which LLMs generate SQL; Kanchinadam *et al.* [7] demonstrated that structurally different variants of the same conceptual model produce divergent answers in Text-to-SQL tasks; and ERBench [8] provided evidence that LLMs commit logical errors when inferring relationships among entities, even in simple schemas. These findings converge on a conclusion: reliable schema generation cannot be delegated to a single prompt; it requires explicit and continuous verification of functional dependencies.

In the multi-agent realm, contemporary evidence confirms that specializing roles reduces logical errors. AttributeForge [9] orchestrates 43 specialized agents to model product schemas in production, while GalaxyWeaver [10] shows that an agent ecosystem can iteratively evaluate a deficient schema and autonomously restructure it until an optimal state is reached. These antecedents suggest that the same strategy can be applied to the relational normalization problem, provided that an agent is devoted to auditing the mathematical dependencies that underlie the schema.

This article presents an empirical framework for autonomous data modeling composed of specialized agents: an analyst agent extracts entities, attributes, and relations from the requirements text; a generator agent builds the data dictionary; a coding agent authors the DDL code for PostgreSQL; and an orchestrator agent evaluates the functional dependencies and iterates over the schema until compliance with 3NF is demonstrated before the final DDL is emitted. The contributions of the article are: (i) the definition of a multi-agent architecture with separated cognitive roles for requirements extraction, documentation, and coding, audited by a dependency verifier; (ii) a verification protocol that operates under Codd's formal normalization criteria [2] as the success metric; (iii) an empirical evaluation on complex real-world specifications measuring the system's success rate in producing anomaly-free 3NF schemas; and (iv) a quantitative comparison of design time against the manual process using TOAD Data Modeler.

The remainder of this article is organized as follows. Section II presents the normalization theoretical framework that supports the system. Section III describes the framework archi-

tecture and the interaction protocols among agents. Section IV details the experimental protocol and test cases. Section V analyzes the obtained results, and Section VI provides the conclusions.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

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- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited,

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- There is no period after the "et" in the Latin abbreviation "et al.".
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TABLE I
TABLE TYPE STYLES

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[a]Sample of a Table footnote.



Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

ACKNOWLEDGMENT

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